Power supply project for operational amplifiers

Group 10:

Mohamed Mostafa Abd El-kareem

Nada Ashraf Abd El-Mohsen

Nesreen Ahmed Abdallah

Nour Abd El-Sater Soliman

Yousef Adel Soliman

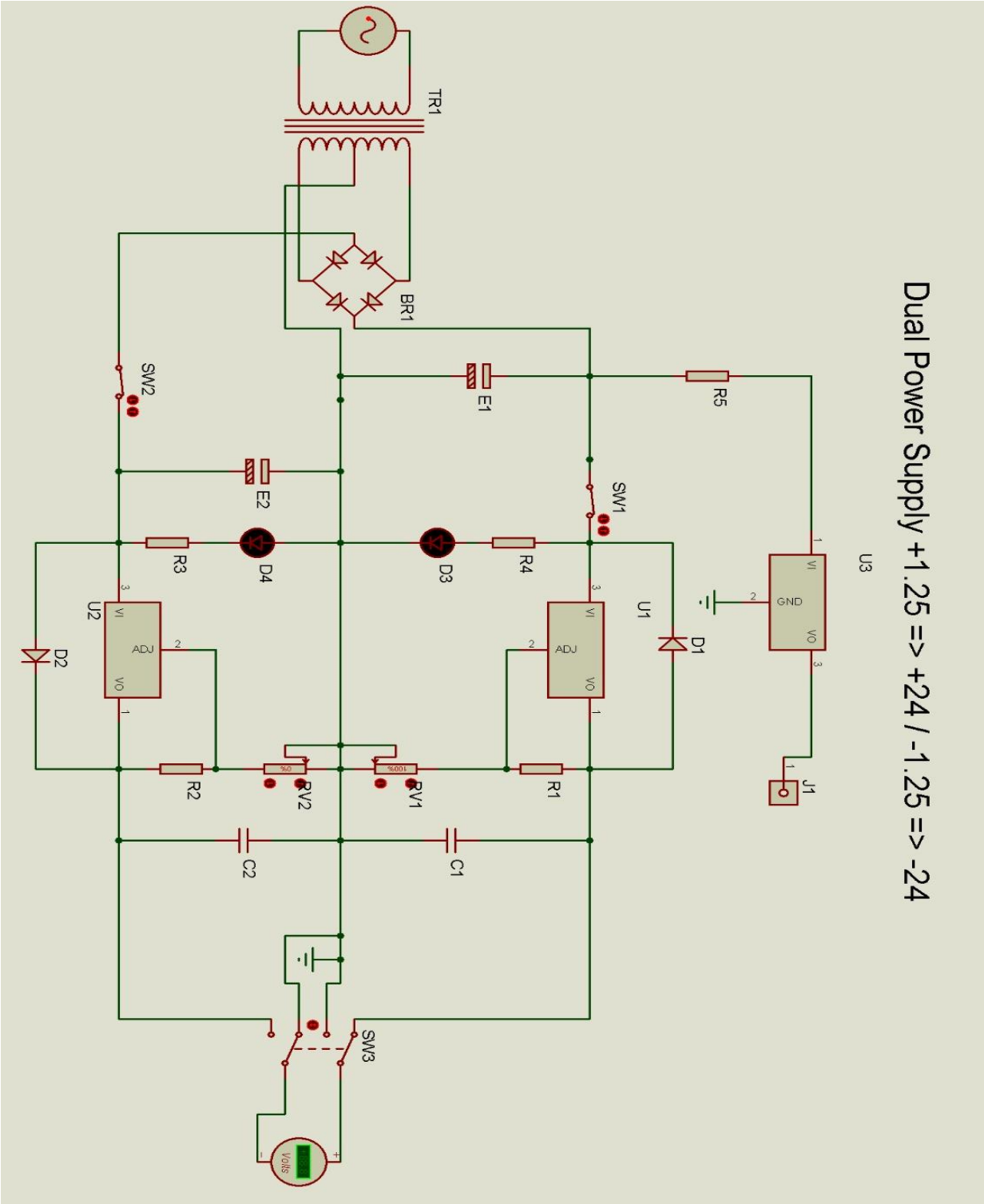
Dr. Eng. :

Emad Zieur

The Content

- Schematic…….2
- Components table……..3
- Project Mechanism……..4
- Final look……5
- The testers……9

- ***Schematic :***


Dual Power Supply +1.25 => +24 / -1.25 => -24
U3
GND
VI
VO
J1
R5
D1
U1
ADJ
R1
SW1
R4
D3
E1
C1
RV1
100%
SW3
Volts
TR1
BR1
SW2
E2
D4
R3
U2
D2
R2
RV2
0%
C2

- **Components table :**

	component	specification	symbol
1	**Transformer**	Centre tapped Step down 24-0-24 (2A)	TR1
2	**Bridge**	full wave rectifier bridge	BR1
3	**Voltage regulator**	LM317	U1
4		LM337	U2
5		LM7805	U3
6	**Resistors**	270 ohm	R1,R2
7		1k	R3,R4,R5
8	**Diode**	1N4007	D1 ,D2
9	**led**	Green	D3
10		Red	D4
11	**Electrolytic Capacitors**	2200uf	E1 ,E2
12	**Ceramic Capacitors**	0.1uf	C1,C2
13	**Potentiometer**	5k ohm	RV1 ,RV2
14	**Switch**	SPST on/off	SW1 ,SW2
15		DPDT on/on Toggle	SW3
16	**Connectors**	1-pin	J1

• Project Mechanism:

Electricity from the wall comes in two flavors: alternating current (AC), constantly switching its direction, and direct current (DC), flowing steadily in one direction. Most electronics, however, crave the consistent flow of DC. That's where the power supply steps in .

1. Transformation:

First up, the transformer adjusts the AC voltage. Imagine gear shifting in a car: high voltage for powerful devices, low voltage for delicate ones. For laptops, the voltage typically drops from 120V or 230V to around 19V.

2. Rectification:

Next, the rectifier acts like a traffic warden, allowing current to flow only one way, converting the AC into a pulsating DC. Think of it as smoothing out the bumpy waves of AC into a steadier stream.

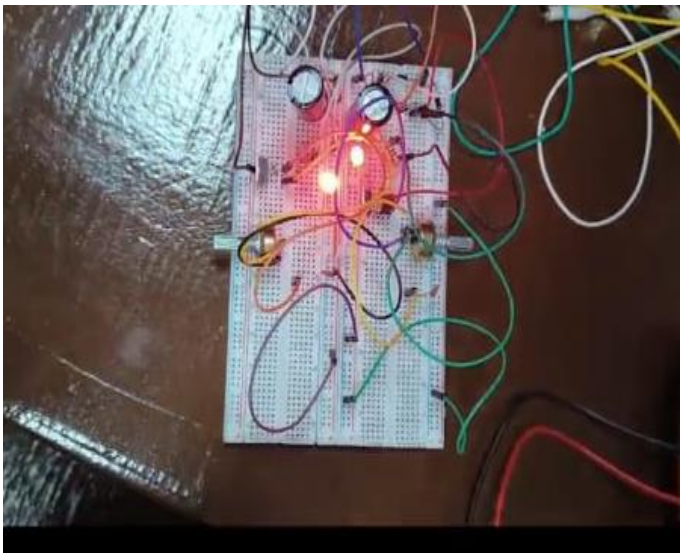
3. Filtration:

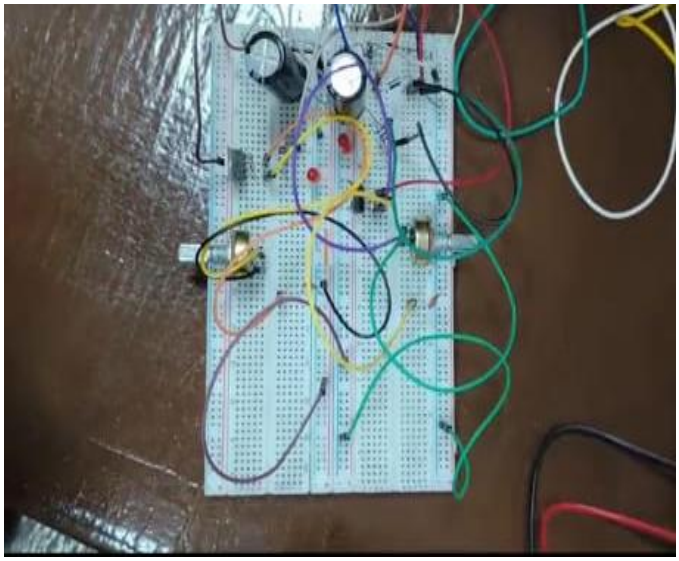
But even that stream isn't perfect. Tiny ripples remain, which can disrupt sensitive electronics. To smooth things out further, filters act like sponges, soaking up the ripples and delivering a cleaner, purer DC flow.

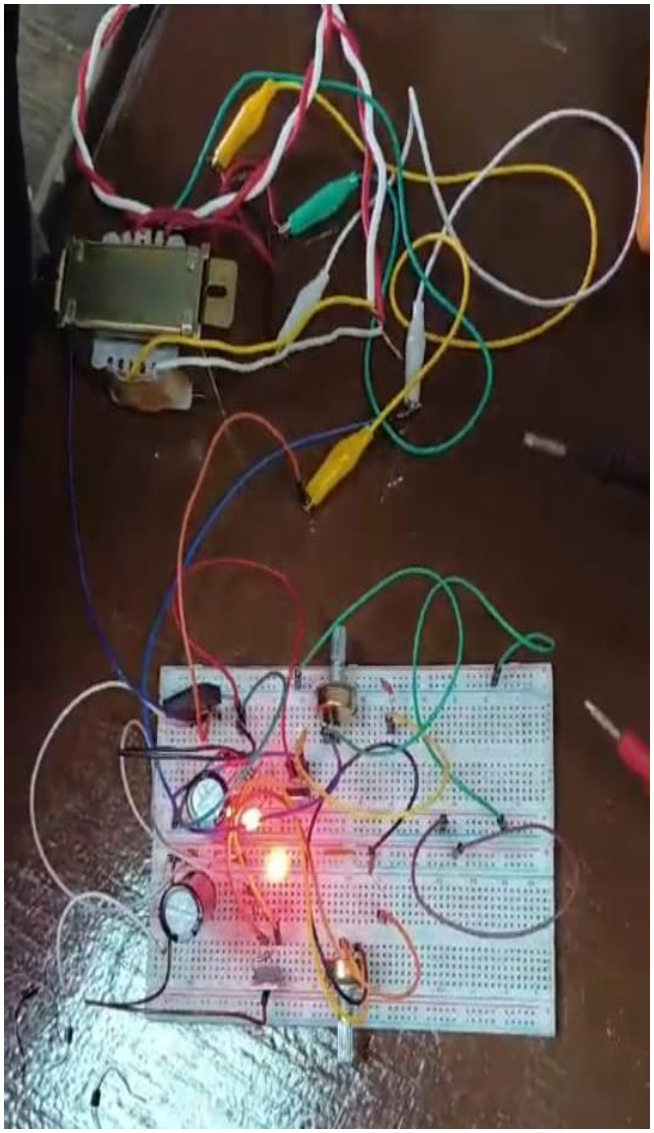
4. Regulation:

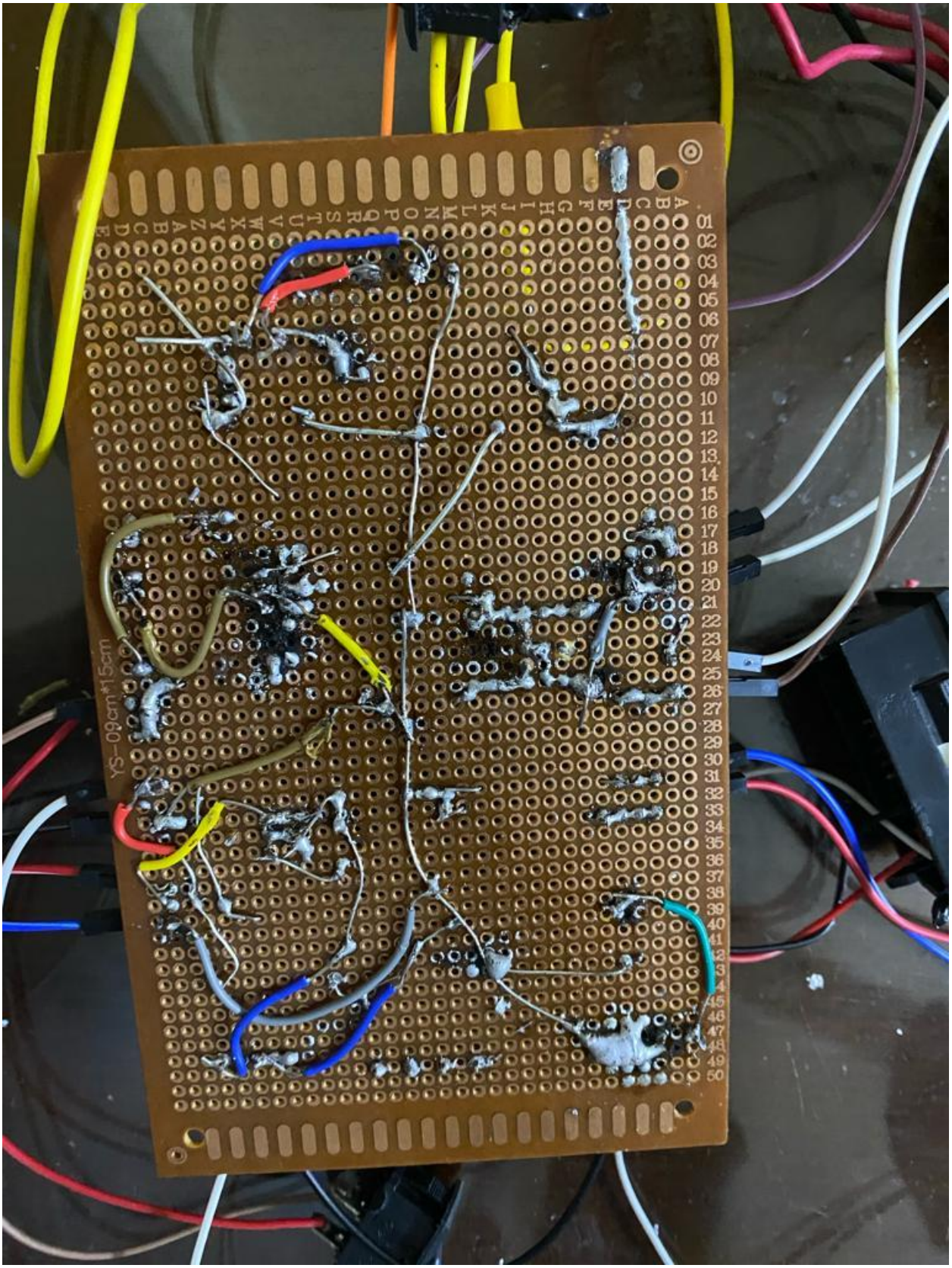
Finally, the regulator acts like a dam, ensuring the voltage stays constant. Imagine a car's cruise control; regardless of the terrain, the speed stays steady. This ensures consistent power for your devices, protecting them from fluctuations that could fry their circuits.

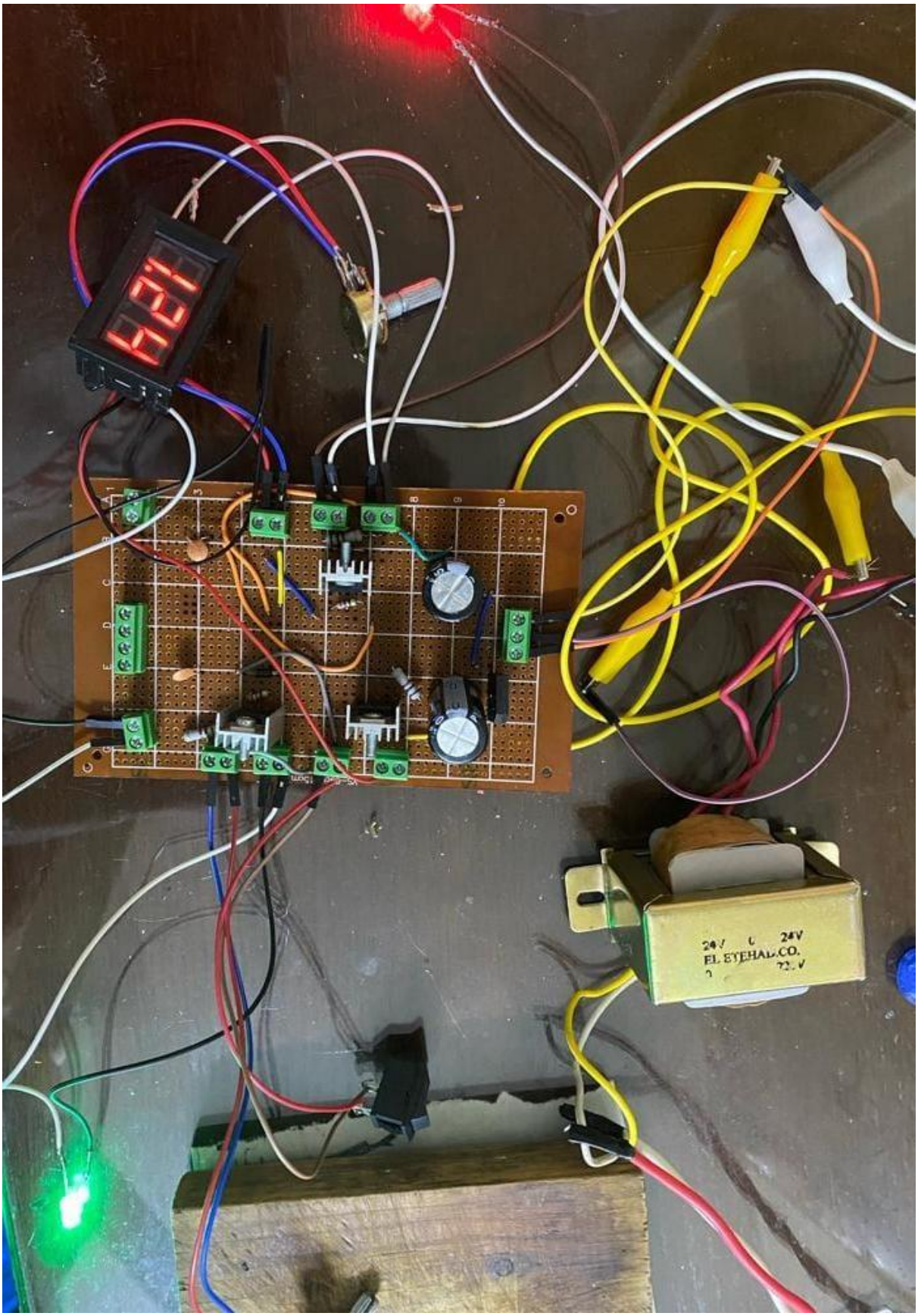
- **Final look :**

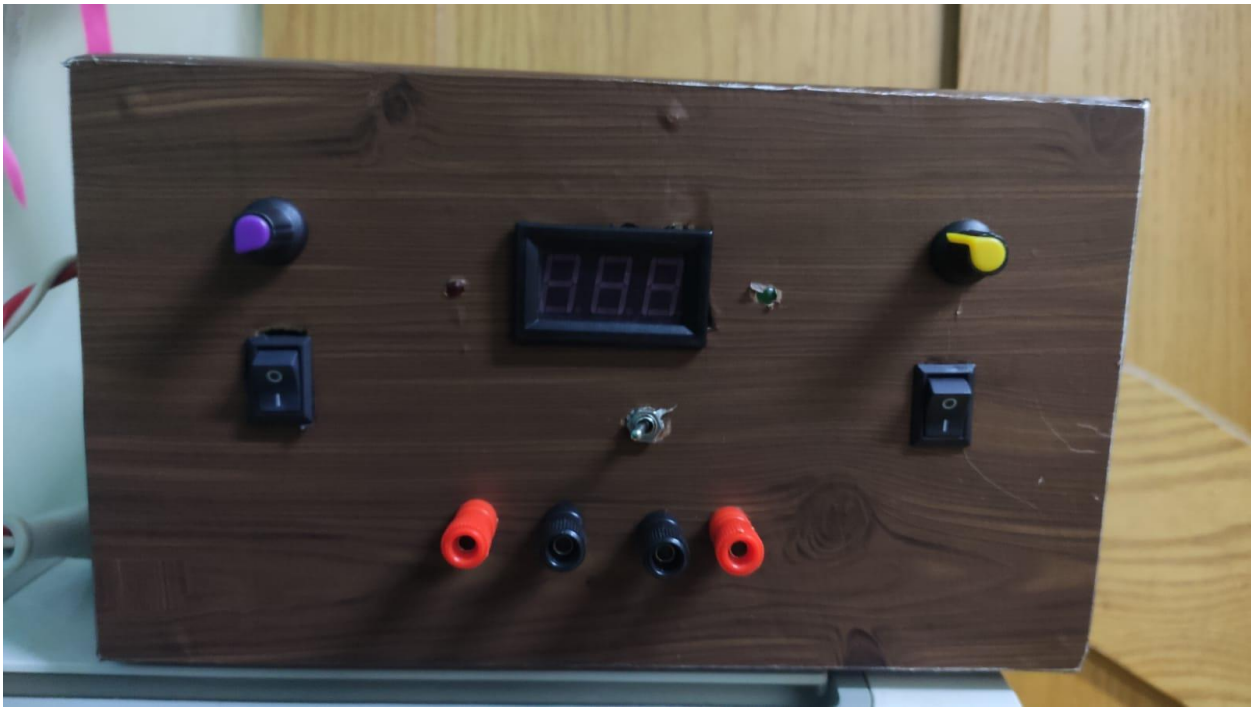


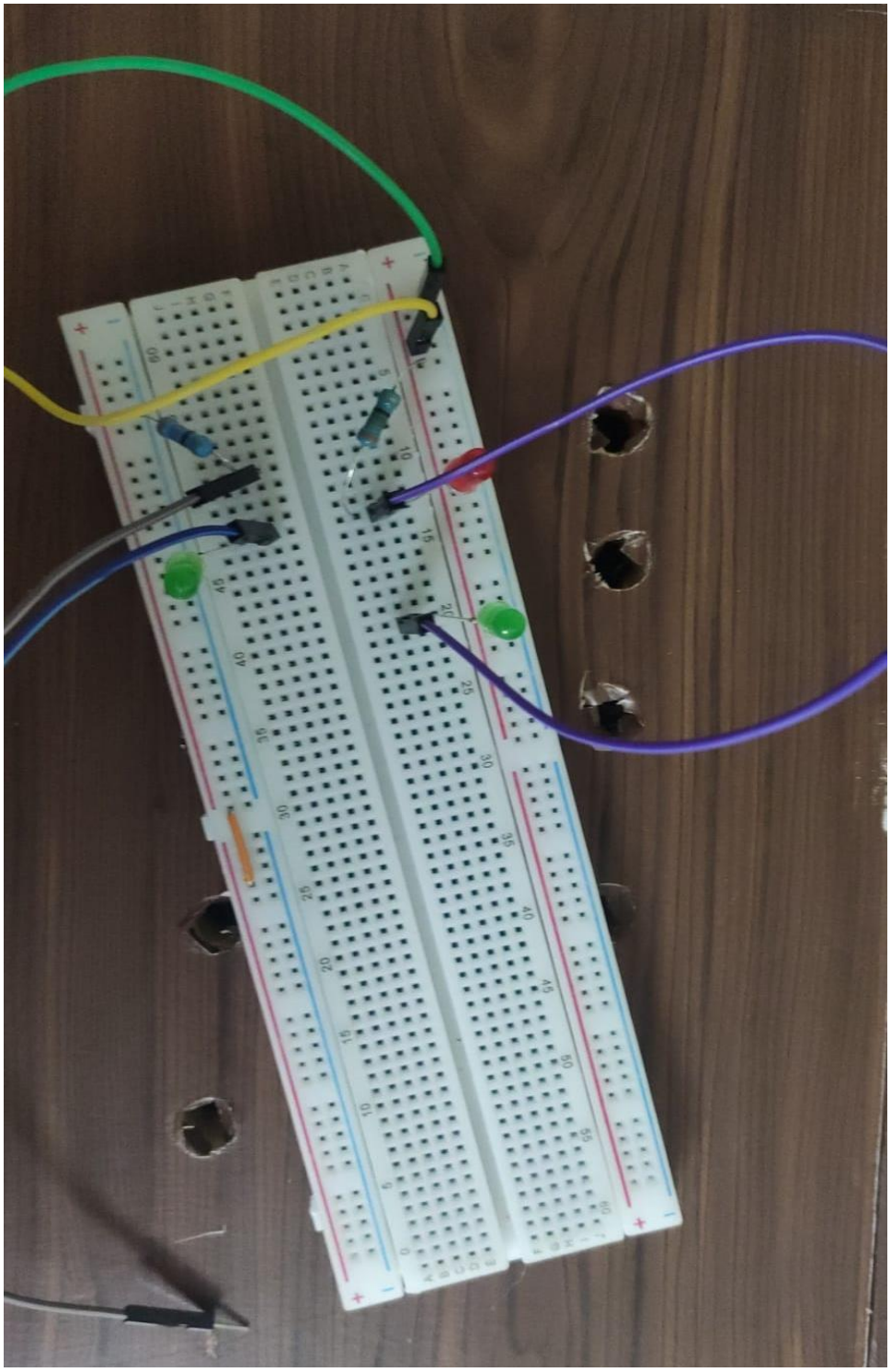





YS-09cm*15cm


24V
24V
EL ETEHAD.CO.



1.20

- **The testers :**